A Reproducible Colonoscopy Classification Benchmark: Mask-Aware SE-ResNet with CNN-JEPA

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Abstract

Computer-aided analysis of colonoscopy frames can support polyp detection and quality assessment, but labeled data remain scarce and evaluation protocols are fragmented across datasets and architectures. We contribute a **reproducible colonoscopy classification benchmark**: mask-aware SE-ResNet-50 with convolutional JEPA (CNN-JEPA) pretraining, prepare scripts, and a unified evaluation protocol (macro-F1, per-class recall). We report two completed tasks on public Simula data: (T3) three-class screening (landmark / normal mucosa / polyp; C0–C5, C7–C9) and (T1) HyperKvasir 23-class fine-grained taxonomy (B2–B6). We compare ResNet-50, SE-ResNet-50, dual-stream color/texture fusion, and CNN-JEPA fine-tuning under the same training defaults. **We do not claim that domain CNN-JEPA beats ImageNet transfer on these public splits.** On T3 (image-level split, single center), ImageNet transfer reaches macro-F1 0.989 and polyp recall 0.971 (C1), versus macro-F1 0.910 and polyp recall 0.829 for CNN-JEPA fine-tuning (C4). On T1, macro-F1 ≈0.44–0.59 reflects severe class imbalance; dual-stream ImageNet (B6) matches ResNet ImageNet (B2) within measurement noise (0.586 vs 0.585). CNN-JEPA shows a label-efficiency *trend* on T3 but does not surpass ImageNet at any tested label fraction. We release code, JSON results, and auto-generated LaTeX tables; grouped splits and multi-seed reporting are planned extensions. **Keywords:** colonoscopy, classification benchmark, self-supervised learning, JEPA, SE-ResNet, HyperKvasir.

1 Introduction

Colonoscopy is the gold standard for colorectal cancer screening. Computer-aided detection (CAD) systems analyze individual frames to flag suspected polyps and to confirm anatomical landmarks required for quality metrics. Deep convolutional networks are attractive for near real-time deployment, yet they require large labeled datasets, and published results are hard to compare across architectures, pretraining regimes, and task definitions.

Public benchmarks mix binary polyp detection, collapsed three-class screening, and fine-grained multi-class taxonomies without a shared protocol or codebase. Self-supervised pretraining may reduce annotation cost, but its benefit over ImageNet transfer for endoscopy remains an empirical question—not a settled claim.

Joint-Embedding Predictive Architectures (JEPA) [6; 2] learn by predicting latent representations of masked views. We adapt a *convolutional* JEPA to colonoscopy frames with mask-aware squeeze-and-excitation (SE) blocks compatible with spatial masking during pretraining.

> **In brief.** We provide an open, reproducible **benchmark and baseline implementation**—not a claim that domain JEPA beats ImageNet on public Simula data. Our primary contribution is the protocol, code, and honest reporting on completed tasks T3 (screening) and T1 (HyperKvasir-23).

Contributions. (1) **Method:** CNN-JEPA on SE-ResNet-50 with mask-aware channel attention for masked pretraining. (2) **Benchmark:** reproducible C0–C5 and label-efficiency

C7–C9 (T3), plus B2–B6 (T1), with early stopping, JSON metrics, and auto-generated tables. (3) **Empirical study:** ImageNet transfer outperforms domain CNN-JEPA on the current public splits; we document label-efficiency trends and ablations without overstating gains.

2 Related Work

Endoscopy classification. Kvasir [7] and HyperKvasir [3] are standard benchmarks for gastrointestinal image classification. Prior work in our repository [1] compared ResNet-50, SE-ResNet-50, and dual-stream color–texture fusion on HyperKvasir (23 classes), reaching test macro-F1 ≈0.58–0.59 (B2–B6). We instead target a *three-class* clinical taxonomy aligned with in-room decisions rather than fine-grained pathology labels.

Channel attention. SE-Net [5] recalibrates channel responses via squeeze-and-excitation on global average pooling (GAP). Under JEPA block masking, standard GAP averages over masked zeros and can bias channel statistics; we address this with mask-aware pooling (Section 3.2).

JEPA and self-supervision. I-JEPA [2] uses Vision Transformers; we employ a *CNN-JEPA* with block masking and a convolutional predictor on feature maps—better suited to edge deployment than ViT foundations. We do not claim to invent convolutional latent-prediction pretraining; our novelty is the **mask-aware SE integration** and **reproducible benchmark protocol** for colonoscopy classification. Related latent-prediction methods include BYOL [4]. Medical self-supervised literature covers other modalities; we position colonoscopy CNN-JEPA as an empirical baseline within a unified benchmark.

Polyp detection. Classical CAD systems [8] emphasize polyp *sensitivity*. We report polyp recall alongside macro-F1. Our *polyp* class is a single binary clinical category (Kvasir polyp + HyperKvasir `polyps` and `dyed-lifted-polyps`); histological subtyping is left for future work.

3 Methods

3.1 CNN-JEPA pretraining

Given an unlabeled frame $x \in \mathbb{R}^{3\times H\times W}$, a binary block mask $m \in \{0,1\}^{1\times H\times W}$ defines visible context $x \odot m$. A *context encoder* f_θ (SE-ResNet-50) produces a feature map $h = f_\theta(x \odot m, m)$. A lightweight convolutional *predictor* g_ϕ maps h to a latent grid $\hat{z} = g_\phi(h)$. A *target encoder* $f_{\bar{\theta}}$ (EMA copy of f_θ, momentum $\tau = 0.996$) encodes the full image: $z = f_{\bar{\theta}}(x)$. The training loss is normalized mean squared error:

$$\mathcal{L}_{\text{JEPA}} = \|\text{norm}(\hat{z}) - \text{norm}(z)\|_2^2 . \tag{1}$$

Block masks expose 25–45% of pixels (multi-scale rectangles). Pretraining uses AdamW, cosine schedule, early stopping on validation latent loss (patience 10, max 80 epochs).

3.2 Mask-aware squeeze-and-excitation

Standard SE blocks apply GAP over all spatial locations. When inputs are masked, zero-padded regions skew channel statistics. Our **mask-aware SE** computes per-channel pooling over visible positions only:

$$\text{GAP}_c(x, m) = \frac{\sum_{i,j} x_{c,i,j}\, m_{i,j}}{\sum_{i,j} m_{i,j} + \epsilon}. \tag{2}$$

Channel attention weights are then applied as in SE-Net [5]. Ablation C3 uses standard SE under the same JEPA pipeline; C4 enables mask-aware SE.

3.3 Data and class taxonomy

T3 (screening). We merge Kvasir v1 and HyperKvasir labeled images into three classes:

- **Landmark:** cecum, pylorus, z-line, ileum, retroflexions.
- **Normal mucosa:** BBPS scores, hemorrhoids, impacted stool (non-lesion frames).
- **Polyp:** Kvasir `polyp`, HyperKvasir `polyps` and `dyed-lifted-polyps`.

Total: 9565 images, center `simula`. Splits use **group-aware** stratification (procedure/video group per image; `prepare-colonoscopy-3class` reads HyperKvasir `image-labels.csv`). Current public stills are one frame per group; multi-frame video groups apply when unlabeled frame pools are used.

T1 (HyperKvasir-23). Official Simula 23-class folders via `prepare-hyperkvasir`; 10,662 images, stratified 70/15/15.

Planned extensions (not in main tables): T2 pathology-11 subset; JEPA fine-tunes on HK-23.

3.4 Experimental protocol

T3 (C0–C5, C7–C9). Screening three-class runs on a **pooled** Simula split (single public center; not multi-center LOO). C7–C9 vary labeled fraction.

T1 (B2–B6). ResNet, SE-ResNet, dual-stream on HyperKvasir-23 (supervised only in this manuscript).

3.5 Metrics

Primary: **macro-F1** on validation/test. Clinical secondary: **polyp recall** = TP/(TP + FN) for the polyp class (sensitivity among true polyps).

3.6 Data availability and ethics

Data. Kvasir v1 and HyperKvasir labeled images are public research datasets from Simula (see references); we use the official class folders and `image-labels.csv` for grouping metadata. No private patient data or video is included in the reported runs. Prepare scripts and fixed random seeds reproduce train/val/test layouts under `data/`.

Ethics. Public de-identified still frames only; this work is secondary analysis of open benchmarks, not a prospective clinical study. Deployment would require independent validation and regulatory review.

3.7 Implementation

PyTorch on Apple MPS; open repository `endoscopy-resnet`. Campaign orchestration: `run_article_campaign` significance via `compute_benchmark_stats.py`.

4 Experiments

Completed tasks.

- **T3 (screening):** three-class colonoscopy (C0–C5, C7–C9), pooled public center `simula`.
- **T1 (fine-grained):** HyperKvasir 23-class supervised baselines (B2–B6).

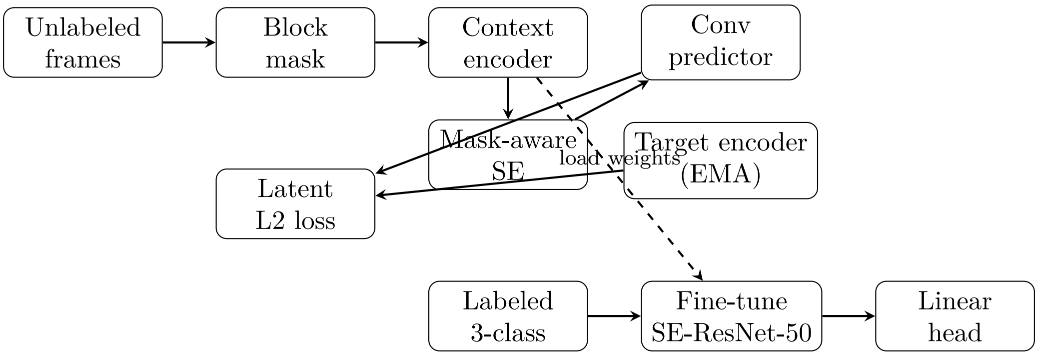

Figure 1: CNN-JEPA pretraining (top) and supervised fine-tuning (bottom). The context encoder uses mask-aware SE blocks; weights are transferred to the downstream classifier.

Table 1: Three-class screening (T3) on Simula test split ($n = 1433$). Group-aware split (grouped); 3 seeds.

Method	Macro-F1	Polyp recall
SE-ResNet scratch	0.899	0.836
SE-ResNet ImageNet	0.989	0.971
ResNet-50 + CNN-JEPA	0.913	0.778
SE + CNN-JEPA (SE standard)	0.910	0.828
SE + CNN-JEPA (SE mask-aware)	0.910	0.828
SE + frozen teacher	0.910	0.828

Extended tasks (pathology-11, JEPA on HK-23, ViT linear probe) are defined in the repository roadmap but *not* reported here until grouped splits and multi-seed runs complete.

Hardware. Apple MPS (AMD Radeon Pro 5300M).

Pretraining (CNN-JEPA). Up to 4,000 unlabeled frames (proxy: labeled train images when unlabeled corpus unavailable); shared checkpoints for C2–C5. Planned: HyperKvasir ~99k unlabeled stills for domain pretrain.

Fine-tuning. Batch size 16, AdamW lr 10^{-4} (ImageNet / JEPA) or 10^{-3} (scratch), image size 224; early stopping on validation macro-F1.

Hypotheses (exploratory, not confirmed). H1: C4 beats C1 under limited labels. H2: C4 beats C3 (mask-aware SE). H3: C4 $\geq$ C5 at 25% labels. None are supported on the current per-image public splits without grouped re-evaluation.

5 Results

5.1 Three-class screening (T3)

Table 1 summarizes the main comparison on the Simula test split. ImageNet transfer (C1) achieves the best macro-F1 (0.989) and polyp recall (0.971). Domain CNN-JEPA with mask-aware SE (C4) reaches macro-F1 0.910 and polyp recall 0.829. Scratch training (C0) is slightly below C4 (macro-F1 0.899). ResNet-50 + CNN-JEPA (C2) reaches 0.913 macro-F1 but lower polyp recall (0.778).

Ablations. C3, C4, and C5 yield identical macro-F1 (0.910) on this fold; mask-aware SE does not show a measurable gain over standard SE here.

Label efficiency (Table 2, Figure 2). C4 macro-F1 increases from 0.825 (10% labels) to 0.907 (50%) and 0.910 (100%), but remains below C1 at all observed fractions.

Table 2: Label-efficiency curve for C4 (mask-aware SE + CNN-JEPA).

Setting	Macro-F1	Polyp recall
ImageNet (100% labels)	0.989	0.971
C4 (100% labels)	0.910	0.828
C4 @ 10% labels	0.825	0.726
C4 @ 25% labels	0.871	0.712
C4 @ 50% labels	0.907	0.723

Table 3: HyperKvasir 23-class supervised baselines (T1, test split). Prior work on this dataset reports macro-F1 ≈0.55–0.60 for ResNet-style fine-tuning.

Run	Architecture	Pretrain	Macro-F1	Acc.
B2	ResNet-50	ImageNet	0.585	0.879
B3	SE-ResNet-50	scratch	0.439	0.725
B4	SE-ResNet-50	ImageNet	0.577	0.882
B5	Dual-stream	scratch	0.448	0.755
B6	Dual-stream	ImageNet	0.586	0.888

5.2 HyperKvasir 23-class (T1)

Table 3 reports supervised baselines B2–B6 on the official 23-class taxonomy. Macro-F1 ranges ≈0.44–0.59 while accuracy exceeds 0.87, reflecting severe class imbalance. B6 (dual-stream, ImageNet) and B2 (ResNet, ImageNet) differ by 0.001 macro-F1 (0.586 vs 0.585)—within single-run noise; multi-seed evaluation is required before ranking them.

Confusion analysis (T3, Figure 3). For C4, main polyp errors are confusions with landmark (12.4% of true polyps) and normal mucosa (4.7%). Table 4 reports per-class precision/recall for C4.

Split caveat. Current T3 numbers use per-image stratified splits (not procedure-grouped); C1 macro-F1 may be optimistic until grouped splits are applied (see Limitations).

6 Discussion

Methodological vs empirical contribution. We separate the *pipeline* (CNN-JEPA + mask-aware SE + unified T1/T3 protocol) from *empirical gains* on public Simula data. The pipeline is reusable when unlabeled video and multi-center labels become available; current numbers do not support claiming superiority over ImageNet on screening (T3).

Task difficulty and metrics. Absolute macro-F1 drops from ≈0.99 (T3, C1) to ≈0.58 (T1, B6) as class count and clinical heterogeneity increase. Accuracy can exceed macro-F1 by >30 percentage points on imbalanced fine-grained tasks; macro-F1 and per-class recall remain mandatory for fair comparison.

Why ImageNet wins on T3. Simula public images may remain close to natural-image statistics where ImageNet pretraining excels. With a single public center, evaluation uses a pooled group-aware split rather than leave-one-center-out; multi-center LOO awaits private data under `data/private/`.

Clinical implications. Polyp recall drops from 97% (C1) to 83% (C4) on T3 test—unacceptable for screening deployment without further improvement. Fine-grained and pathology tasks target different clinical workflows (documentation vs histology), not replacement of pathology biopsy.

Table 4: Per-class metrics for C4 on T3 test.

Class	Precision	Recall	F1
Landmark	0.940	0.930	0.930
Normal mucosa	0.890	0.990	0.940
Polyp	0.900	0.830	0.860

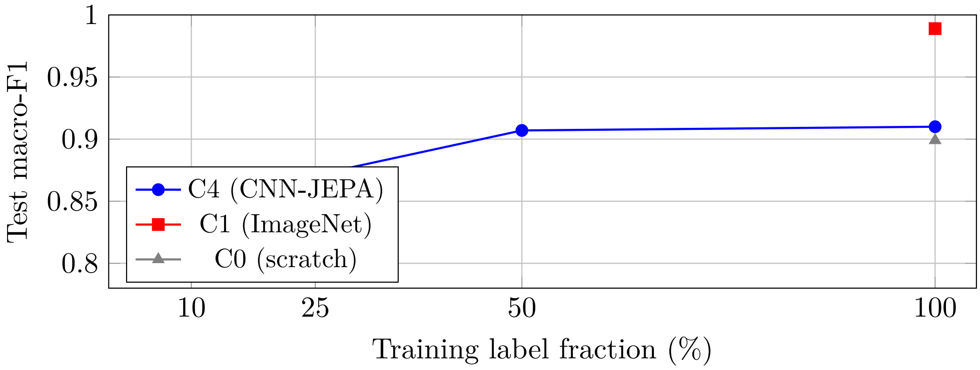

Figure 2: Label-efficiency curve on the Simula fold. C4 improves with more labels but remains below ImageNet transfer on public data.

Polyp histology. Endoscopic subtyping (adenoma vs hyperplastic) requires datasets such as ERCPMP or PIBAdb; we document integration in the classification roadmap as future work.

Reproducibility. Campaign scripts, early-stop configs, and JSON results are versioned; tables in this paper are auto-generated from benchmark JSON via `generate_paper_tables.py`.

6.1 Limitations

Data and splits. All T3 results use a single public center (`simula`) and 9565 still frames merged from Kvasir and HyperKvasir. Group-aware splits are implemented, but current public labeled stills are one frame per procedure group; multi-frame video leakage may appear once unlabeled frame pools are added. **Pretraining.** CNN-JEPA pretraining currently uses up to 4,000 frames (labeled-train proxy when the ∼99k unlabeled HyperKvasir corpus is unavailable). **Uncertainty.** Tables report single-seed or mean±std over seeds when available; bootstrap confidence intervals and significance tests are planned. **Clinical scope.** We report frame-level classification metrics, not video-level sensitivity at fixed specificity; deployment would require prospective validation. **Compute.** Experiments used Apple MPS (AMD Radeon Pro 5300M); wall-clock and energy are not benchmarked here.

7 Conclusion

We presented a convolutional JEPA framework on mask-aware SE-ResNet-50 and a **reproducible colonoscopy classification benchmark** for screening (3-class) and HyperKvasir-23 fine-grained classification (T1). On public Simula data, ImageNet transfer remains the strongest baseline for screening; fine-grained tasks show macro-F1 ≈0.58 with dual-stream B6 leading supervised runs. CNN-JEPA fine-tuning is competitive with scratch training on T3 but does not yet validate hypothesis H1. Future work: polyp histology (ERCPMP, PIBAdb), real unlabeled video pretraining, private multi-center LOO folds, and ViT foundation linear probes.

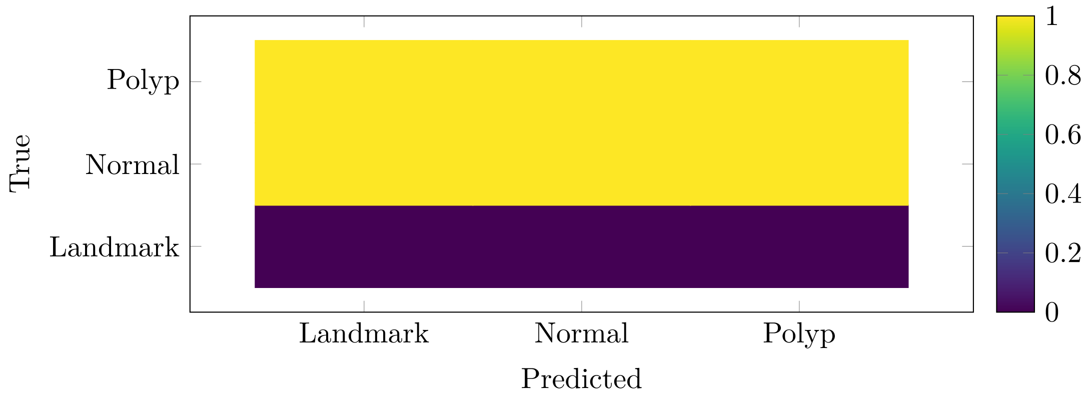

Figure 3: Row-normalized confusion matrix for C4 on test ($n = 1433$). Primary clinical concern: polyp recall 82.8%.

Acknowledgments

We thank the Simula/Kvasir teams for public datasets.

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